

De-Ess

A split-band or wideband de-esser that turns sibilance down only when it gets too loud.



Overview

De-Ess listens only to the esses - a band around **FREQ**, or everything above it - and when that band goes over the threshold, it turns the sibilance down, by no more than **RANGE**. It reacts very fast (about 0.3 ms), so even short, sharp esses are caught.

In **SPLIT** mode only the esses band is turned down and the rest of the voice is untouched, which is the most transparent way to work. In **WIDE** mode the whole signal ducks on each ess, like the classic hardware boxes. One detector serves both channels, so the stereo image never shifts.

Reach for it for

- Sibilant lead and backing vocals
- Spoken word, voice-over and podcasts
- Fret squeak and harsh pick noise on acoustic guitar
- Splashy hi-hats and cymbals in a drum overhead (**HIGH** shape)
- Taming a bright vocal after heavy compression or EQ

Controls

Mode and shape

SPLIT	Only the esses band is turned down; the rest of the voice is untouched. Transparent - the default.
WIDE	The whole signal ducks on each ess, like the classic hardware boxes. Can sound more natural on some voices, but is easier to hear working.
BAND	The detector listens to a band around FREQ , and in SPLIT mode a dip around FREQ is what turns down. Best for a voice whose esses sit in one place.
HIGH	The detector listens to everything above FREQ , and in SPLIT mode a high shelf dips everything above FREQ . Best for broad, airy sibilance and cymbals.
LISTEN	Hear what the detector hears - just the esses band. Use it to tune FREQ onto the esses, then switch it off.

Knobs

Each knob's value shows while you hover over it or drag it; while dragging, the display in the top strip also shows the control's name and value.

FREQ	Where the esses are, 2 - 12 kHz (starts at 6.5 kHz). Lower for male voices, higher for female voices and cymbals.
THRESHOLD	The level of the esses band where De-Ess starts working, -50 to 0 dB. Turn it down for more de-essing. Above the threshold it works at about 5:1, with a soft 6 dB knee.
RANGE	The most it will ever turn the esses down, 0 - 24 dB. Keeps it from lisping on loud esses.
RELEASE	How quickly the esses band comes back after each ess, 20 - 300 ms.
OUTPUT	Output level, -12 to +12 dB.

Lamp and meters

ESS	The teal lamp lights while De-Ess is turning the esses down (by more than half a dB).
ESS LV	The level of the esses band (bottom -60 dB, top 0 dB). The red line marks THRESHOLD: when the lights pass it, De-Ess is working.
GR	How much the esses are being turned down, lit from the top, up to 24 dB.

The top strip

Preset display	Shows the preset number and name. Click it to open the preset list.
< and >	Step to the previous or next preset.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper at -0.3 dBFS (no latency), or a lookahead limiter that holds the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). CLIP is on by default; its key flashes red while it is clipping.
S / M / L	Panel size: small, medium or large. A new De-Ess opens at L. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

De-Ess has no input stage: there is no TRIM, GUARD or SAT.

Presets

A preset sets the mode, shape, FREQ, THRESHOLD, RANGE and RELEASE, and switches LISTEN off. OUTPUT and the OUT stage are left as they are.

Init	The starting point: SPLIT, BAND at 6.5 kHz, -26 dB threshold, 10 dB range, 60 ms release.
Female Vocal	SPLIT, BAND at 7.5 kHz with a quicker 50 ms release.

Male Vocal	SPLIT, BAND at 5.5 kHz, slightly gentler, 70 ms release.
Gentle Air	SPLIT, HIGH above 9 kHz with only 6 dB of range - softens the top without dulling it.
Hard Esses	WIDE, BAND at 6.5 kHz with a low threshold, 16 dB of range and a fast release, for very sharp sibilance.
Acoustic Guitar	SPLIT, BAND at 4.5 kHz, 6 dB range and a slow release, for string and pick noise.
Hats and Cymbals	SPLIT, HIGH above 10 kHz, 8 dB range.
Podcast Voice	SPLIT, BAND at 6 kHz with a low threshold and 12 dB of range for steady spoken word.

Tips

- Switch LISTEN on and sweep FREQ until the esses sound loudest and harshest, then switch LISTEN off and set THRESHOLD.
- Watch the ESS LV meter: set THRESHOLD so only the esses cross the red line, not the rest of the words.
- If the voice starts to lisp, lower RANGE rather than raising THRESHOLD.
- Put De-Ess after compression and bright EQ, which both tend to bring the esses up.
- Use SPLIT for transparency; try WIDE when SPLIT makes the esses sound dull or hollow.

File: gfx-de-ess.jsfx